

Esercizio 1. (Alessandro Martina) Prove that G -syndetic sets are weakly G -thick.

Esercizio 2. (Francesco Sulpizi) Assume that $\mathcal{X}$ is $\Delta(\mathcal{Z})$ -type-definable and G acts transitively on $\mathcal{X}$. Let $\mathcal{Y} \subseteq \mathcal{X}$ be a $\Delta(\mathcal{Z})$ -type-definable set with a small G -orbit. Prove that $\mathcal{Y}$ is definable by a G -syndetic type.

Esercizio 3. (Leonardo Centazzo) Prove that the following are equivalent

1. $\mathcal{D}$ is weakly G -thick
2. $\mathcal{D} = \mathcal{C} \cap \mathcal{B}$ for some G -syndetic set $\mathcal{B}$ and some G -thick set $\mathcal{C}$
3. there is a non G -syndetic set $\mathcal{C}$ such that $\mathcal{D} \cup \mathcal{C}$ is syndetic.

Esercizio 4. (Roberto Carnevale)

Esercizio 5. (Matteo Bisi) Let $\mathcal{D}$ be a $\Delta(\mathcal{Z})$ -type-definable set. Are the following equivalent?

1. $\mathcal{D}$ is G -thick
2. $\mathcal{D}$ is definable by a G -thick type.

Esercizio 6. (Davide Peccioli)

Esercizio 7. (Pietro Giura) Let $\mathcal{X} = \mathcal{Z} = \mathcal{U}$. Prove that if $p(x) \in S(\mathcal{U})$ is finitely satisfiable in every $M \supseteq A$ then it is thick under the action of $G = \text{Aut}^f(\mathcal{U})$. Is the same true for incomplete types?

Esercizio 8. (Costanza Furone) Prove that G -syndetic sets are weakly G -thick.

Esercizio 9. (Angela Dosio)